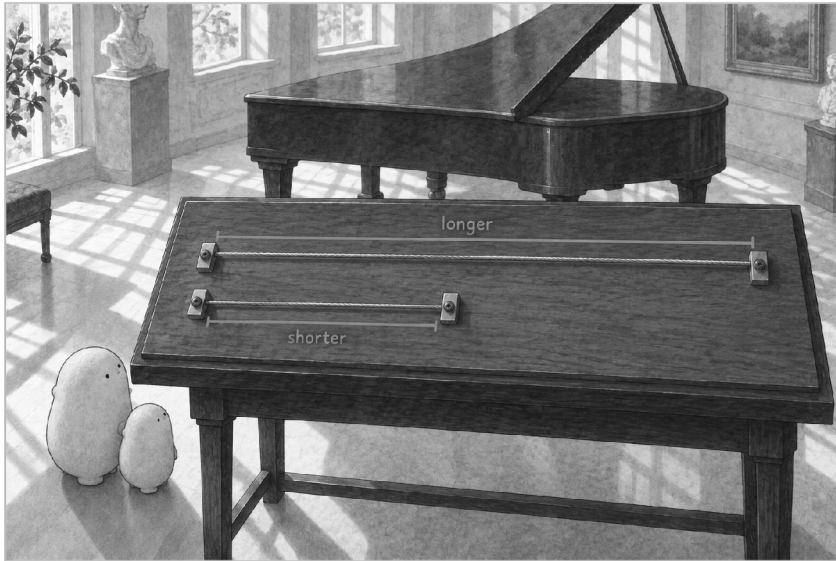


Longer by how much?

The note inside. · Curiosity-led family day

Name _____

Date _____



PRETEND DRAWING LENGTHS · CENTIMETERS

Line	A	B
Length (cm)	24	9

These are invented drawing lengths. Do not measure the painting or real piano strings.

A comparison model, not two measured piano strings.
Compare length while imagining the material, thickness, and tension stay the same.

1. Line A is 24 cm long. Line B is 9 cm long. How much longer is A?

$$24 - 9 = \underline{\quad}$$

Draw or write your thinking.

2. Imagine drawing A and B end to end. What is their total length?

$$24 + 9 = \underline{\quad}$$

Draw or write your thinking.

3. Little erases 6 cm from line A. How long is that line now?

$$24 - 6 = \underline{\quad}$$

Draw or write your thinking.

Sketch two bars that start at the same place. Show which part is the extra 15 cm. A sketch need not be actual size.

GROWN-UP NOTE About 15 min: 3 to read the table, 9 to sketch or calculate, 3 to explain. A centimeter is a unit of length; cm is its short name. Use number-line hops or bar sketches, and accept a spoken strategy. The table supplies all numbers. These lengths do not predict exact piano notes. No instrument or measuring tools are needed.

The note inside

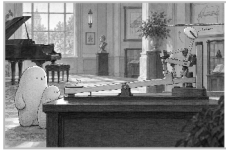
The note inside. · Curiosity-led family day

Name _____

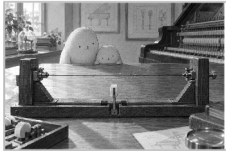
Date _____



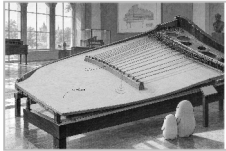
One key sinks beneath Little's hand. A note rings through the museum. "Was it hiding in there?" Little asks. Big listens beside the bench. The key hardly moved. Something inside the piano must have done much more.



An enormous model gives them a peek. A key tips; connected parts lift a felt-covered hammer. Tap! The hammer strikes a string and comes away. "A tiny hammer?" Little whispers. This team of moving parts is called the action.



The hammer has fallen back, but the string is not still. It quivers: back and forth, back and forth. That quick motion is vibration. "The tap is over," Little says, "but the moving isn't!" How can such a thin string sound so big?



Little follows a string to a wooden bridge. The bridge carries vibration into a broad soundboard. That wood vibrates too, moving more air. The air carries sound to listening ears. "The wood joins in!" Little says. A thin string has a roomy helper.



Two strings stretch across another model. One is longer. If everything else stays the same, the longer string vibrates more slowly and sounds lower. The shorter one sounds higher. "Different voices!" says Little. How high or low a note sounds is its pitch.



With the pedal up, Little releases a middle key. A felt damper hushes its strings. Little sketches key, hammer, strings, wood. "The note wasn't hiding. We set things moving!" Big leaves room for one more why.

Talk together: What did Little first guess about the note? Which moving parts helped change that guess? How does vibration get from a string to the air? Name the two wooden helpers.

GROWN-UP NOTE About 15 min: share the six short scenes, then revisit a picture. Read aloud for Leo; pointing, talking, and pausing count. Let Little's first guess hang in the air before following the clues. The foldable carries the same story.

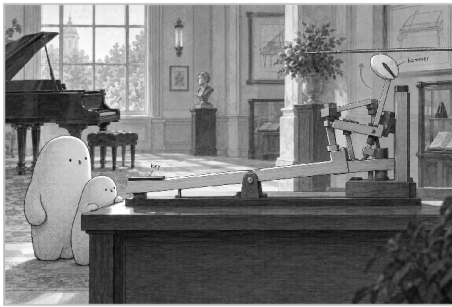
Give the note a path.

The note inside. · Curiosity-led family day

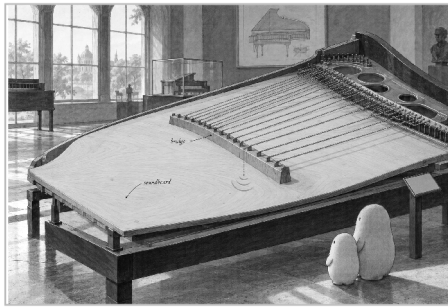
Name _____

Date _____

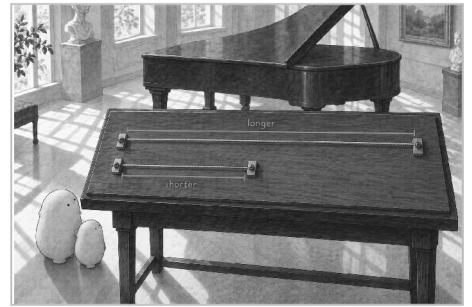
Choose one picture from Little's search. Tell what moves and what happens next. Draw or write a clue that helped you understand the note.



A hidden tap



The wood joins in



Two lengths to notice

Draw one detail from your scene. Add an arrow or a label to show a movement or a difference.

key · hammer · string · vibration · bridge · soundboard · pitch

When a key moves, ... What does the action help the hammer do?

The soundboard helps because ... How does the broad wood help move air?

If only the string's length changes, ... Which length could make the lower sound?

Read it together and point to the detail your words explain. If you are writing, check the first capital and the ending mark.

GROWN-UP NOTE About 15 min: 3 to choose and talk, 4 to draw, 6 for 1–3 sentences, 2 to read back. One careful idea is enough. Welcome speech, dictation, invented spelling, or a labeled drawing. Write Leo's words as he says them, then read them back together.